

Sri Lanka Institute of Information Technology

B. Sc. (Hons) in Information Technology
Specializing
Data Science

Final Examination
Year 3, Semester I (2019)

IT3031-Database Systems and Data-driven
Application

Duration: 2 Hours

June 2019

Instructions to Candidates:

- ◆ This paper is preceded by a 10-minute reading period. The supervisor will indicate when answering may commence.
- ◆ Write answers in the booklet given.
- ◆ The paper contains 4 questions. Answer All Questions.
- ◆ Total marks for the paper is 100.
- ◆ The paper contains 6 pages with the Cover Page.
- ◆ Electronic devices capable of storing and retrieving text, including calculators and mobile phones are not allowed.

Question 1

(25 marks)

Consider the following XML document for the questions given below:

```
<?xml version="1.0" encoding="UTF-8"?>
<dept bldg = "101">
  <employee id="135">
    <name>Ramesh</name>
    <phone>0714123123</phone>
    <office>344</office>
    <age>38</age>
  </employee>
  <employee id="137">
    <name>Malith</name>
    <phone>0776126126</phone>
    <office>345</office>
    <age>45</age>
  </employee>
  <employee id="138">
    <name>Prabash</name>
    <phone>0777123123</phone>
    <office>345</office>
    <age>41</age>
  </employee>
</dept>
```

Note that the above XML document stored in *department* (*id int, emp xml*) table created in MS SQL Server and it contains only single record.

1. Compare XML data model verses the relational data model. (4 marks)
2. Write an XQuery (FLWR) expression to retrieve the *name* of the employee whose employee id 137. (3 marks)
3. Delete the employee details of *Prabash* from XML database. (3 marks)
4. What would be the output of the following XQuery expression?

```
SELECT xText.query ('
  let $emp := /dept/employee
  return <results>
    {
      for $x in $emp
      where $x/age>=40
      order by ($x/age)[1]
      return ($x/name, $x/phone)
    }
  </results>
')
```

FROM department

(4 marks)

5. Write an expression to insert the following employee details after employee ID 135.

```
<employee id="136">
  <name>Samanthi</name>
  <phone>0714123222</phone>
  <office>340</office>
  <age>21</age>
</employee>
```

(5 marks)

6. Write an XQuery expression to extract data from employee xml document and create an HTML table containing the names of all the employees along with their respective contact numbers. Use the following HTML tags in your answer.

```
<table>
  <tr>
    <th>Name</th>
    <th>Contact details</th>
  </tr>
  <tr>
    <td>Ramesh</td>
    <td>0714123123</td>
  </tr>
  ... ..
  <tr>
    <td>Prabash</td>
    <td>0777123123</td>
  </tr>
</table>
```

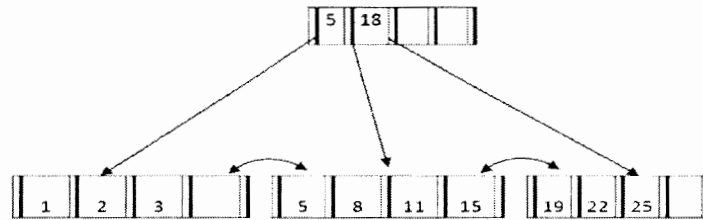
(6 marks)

Question 2

(25 marks)

1. Briefly explain the three main file organizations.
(3 marks)
2. Consider a student relation containing records with student id, name, age and gpa. Assume that the records are stored in a hash file where the hash function is based on the student's age.
"Retrieving records with gpa of 3.0 is inefficient with the above file"
State whether the statement above is true or false with reasons.
(3 marks)
3. Briefly explain two disadvantages that exist in static hashing.
(3 marks)
4. Explain the terms dense and sparse index.
(3 marks)
5. Why can't we create more than one clustered index in a table?
(3 marks)

6 Consider the B+ tree index below.



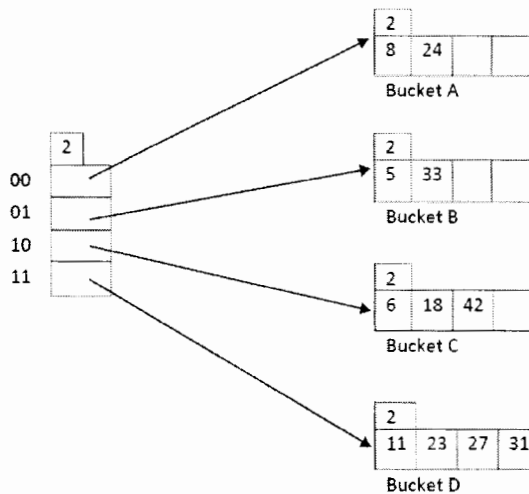
i. Illustrate the tree after inserting 12.

(3 marks)

ii. illustrate the tree after deleting 22 from the original tree.

(2 marks)

7 Consider the Hash index below.



Illustrate the hash index after inserting 55.

(5 marks)

Question 3

(25 marks)

1. List and briefly explain the properties of a transaction. (4 marks)
2. Briefly explain the following terms in database transactions:
Serial Schedule and *Serializable Schedule* (3 marks)
3. Briefly explain the rules in Strict 2 Phase Locking Protocol. (3 marks)
4. List the possible conflicts that may arise with interleaving actions of transactions. (3 marks)

5. Consider the following part of the schedule.

T ₁	T ₂	T ₃	T ₄
S(A)			
R(A)			
	X(B)		
	W(B)		
S(B)			
R(B)			
			S(D)
			R(D)
	X(A)		
	W(A)		
		S(C)	
		R(C)	
			S(C)
			R(C)

Assume that Transaction T₁ is higher priority than transaction T_{i+1} (i.e. transaction T₁ has higher priority than T₂; T₂ has higher priority than T₃; and T₃ has higher priority than T₄).

- How does a deadlock occur? (2 marks)
- How does the DBMS detect a deadlock? (2 marks)
- How the DBMS resolves deadlock (after it has occurred). (2 marks)
- Draw a wait-for graph for the schedule given above to detect the deadlock. (3 marks)
- Draw the schedule again considering deadlock prevention algorithm: *Wait-Die approach* (3 marks)

Question 4

(25 marks)

Consider the following three relations and their associated statistics:

Emp (eno, ename, salary, dept) 10,000 tuples on 1000 pages

Proj (pno, pname, budget, dept) 1000 tuples on 100 pages

Work (eno, pno, sdate, hours) 100,000 tuples on 10,000 pages

The *Emp* table contains tuples of all employees, *Proj* table contains tuples of all projects and *Work* table has tuples indicating employees assigned to projects. Suppose an important query is to find the details of project assignments to employees of a given department using an SQL statement of the form:

```
Select e.eno, w.pno, w.hours
From emp e, work w
Where e.eno = w.eno AND e.dept = 'D3';
```

Assume that project assignments to employees in the Work table are uniformly distributed among all the employees and across all the projects and the employees are uniformly distributed across 200 departments. For convenience, we also assume that all fields are of equal size in each relation. There are 20 buffer pages.

- 1 What is the expected number of tuples in the result of this query?
(3 marks)
- 2 Suppose a selection on Emp (with dept= 'D3') followed by an indexed nested loop join is the preferred method of executing the query. To speed up this query, you are allowed to create an index on each relation. Assuming that index entries are stored in the form <key, rid>, what type of indexes (hash/ B+ tree, clustered/unclustered) would you choose and on which attributes? Give reasons for your choice.
(7 marks)
- 3 Estimate the number of page I/Os for executing the query based on the indexes you have chosen and the corresponding plan.
(7 marks)
- 4 Assume that we want to join two relations *Emp* and *Proj* on *Emp.dept=Proj.dept* conditions. If there were no indexes on the two tables, what would be the cost of a *sort merge join*?
(8 marks)

- End of the Examination Paper -